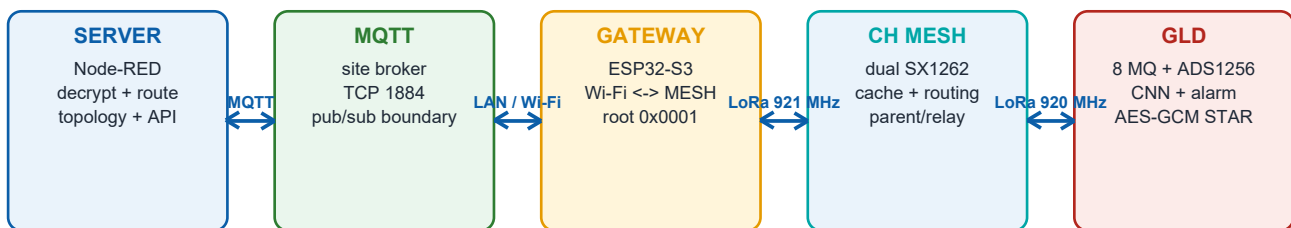


PERTAMINA GLD

Whole-System Architecture

Server, MQTT, Gateway, CH Mesh, GLD and sensor pipeline



Three operational planes

- DATA** (Blue box): Normal telemetry uses CH cache + server pull; alarm uses immediate push.
- CONTROL** (Teal box): Topology discovery, pull route, signed downlink and mode switching.
- MAINTENANCE** (Orange box): Dataset mode uses direct GLD Wi-Fi/MQTT and bypasses CH/Gateway.

DOCUMENT BASIS

Architecture reference derived from the current local repository source and checked-in Node-RED flows on 13 August 2026. It is static evidence, not proof that broker, deployment, database, RF links or devices are currently running.

Document status: Source-based technical reference
Active source versions: GLD 0.8.18 | CH 0.7.3 | Gateway 0.1.5 | Protocol 0.2.0
Repository: D:/Github/PertaminaGLD

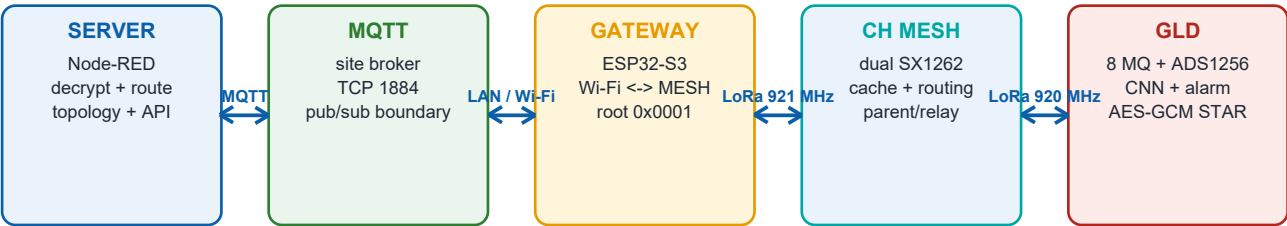
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1. Executive summary

The production architecture is a layered telemetry and control system. The server owns decryption, replay policy, topology and command signing. The Gateway bridges the site IP network to LoRa MESH. Cluster Heads cache GLD records, build routes and schedule downlinks. GLD owns sensing, nulling, inference, physical alarm, encryption and battery power-cycle behavior.



Three operational planes

- DATA** Normal telemetry uses CH cache + server pull; alarm uses immediate push.
- CONTROL** Topology discovery, pull route, signed downlink and mode switching.
- MAINTENANCE** Dataset mode uses direct GLD Wi-Fi/MQTT and bypasses CH/Gateway.

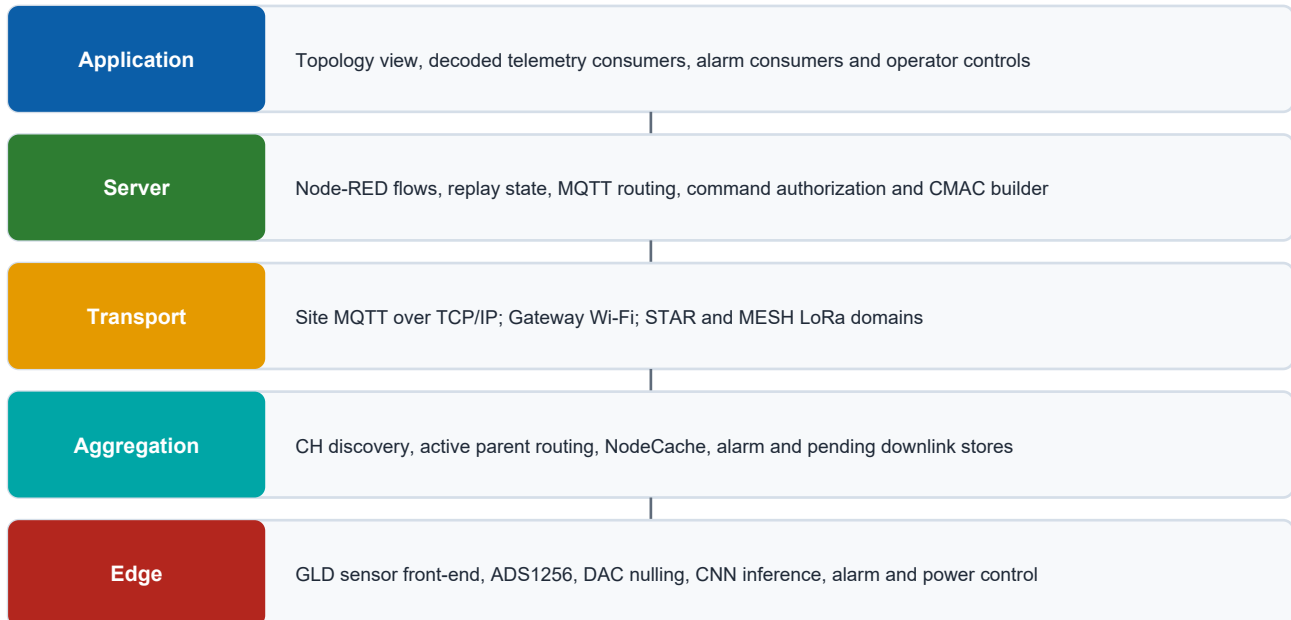
KEY BEHAVIOR

Normal telemetry is cached by CH and retrieved by server pull. Alarm telemetry is pushed immediately with hop-by-hop acknowledgement. Dataset mode is a separate direct GLD-to-MQTT maintenance path.

Component	Current default	Primary responsibility
Server / Node-RED	Broker 127.0.0.1:1884 in flow snapshot	Decode, AES-GCM decrypt, replay checks, topology, command signing
Gateway	ID 0x0001	Wi-Fi/MQTT to LoRa MESH bridge and alarm durability boundary
Cluster Head	ID 0x0010	Dual-radio routing, GLD cache, alarm relay, pull response, downlink scheduler
GLD	ID 0x1001	Eight-sensor acquisition, nulling, CNN inference, alarm and AES-GCM STAR uplink

2. System context and operational planes

The system should be understood as three overlapping planes. Treating every message as a simple continuous uplink hides the most important behavior: normal records are pull-oriented, alarms are push-oriented, and dataset capture bypasses the radio hierarchy.



2.1 Data plane

- Normal GLD records are stored in the owning CH NodeCache and returned only when the server sends a routed pull request.
- Alarm records are pushed toward the root Gateway immediately and retained until the relevant acknowledgement condition is met.

2.2 Control and topology plane

- CH_CONFIG discovery establishes active parents toward the Gateway root.
- CH_HELLO messages populate the server topology state.
- Server node commands are signed, source-routed when required and scheduled by the owning CH.

2.3 Maintenance and dataset plane

- GLD dataset mode enables Wi-Fi and publishes eight-channel records directly to the dataset broker topics.
- Nulling remains local/offline and persists the calibration profile in NVS.

3. Server, broker and Node-RED

3.1 Site broker relationship

Gateway and Node-RED are intended to use the same MQTT broker. Node-RED may use loopback when it runs on the broker host. A remote Gateway must use the server PC LAN address because 127.0.0.1 on the Gateway refers to the Gateway itself.

DEPLOYMENT GAP

Gateway source still contains deployment placeholders for site MQTT host and credentials. The checked-in Node-RED broker snapshot is 127.0.0.1:1884 without TLS. These values describe configuration, not a verified live broker.

3.2 Main MQTT topics

Topic	Direction	Purpose
gld/gateway/uplink	Gateway -> Node-RED	JSON wrapper containing the complete received MESH AppFrame
gld/gateway/topology	Gateway -> Node-RED	CH HELLO and CH_CONFIG topology reports
gld/gateway/status	Gateway -> server/operator	Wi-Fi, MQTT, mesh readiness, queue and identity status
gld/gateway/cmd/pull	Node-RED -> Gateway	requestId and explicit CH hop list
gld/server/cmd/node	Operator/server -> Node-RED	High-level authenticated GLD command request
gld/gateway/cmd/node	Node-RED -> Gateway	Signed node command ready for MESH transport

3.3 Decode and output routing

Output	Meaning
gld/server/decoded	Authenticated non-alarm production telemetry
gld/server/alarm	Authenticated alarm telemetry
gld/server/integrity	Key, tag, schema, frame or replay-state integrity failure
gld/server/replay	Exact encrypted replay or AES-GCM nonce reuse with different content
gld/server/test	Non-production/test-range input separated from production alarms
gld/server/topology	Normalized CH and GLD topology events

3.4 Server state and persistence

- pglTopology stores parents, discovery observations, Gateway link quality, HELLO state and computed routes.
- GLD discovery ownership uses pull request correlation, not merely the last relay source ID.
- Replay state can be persisted atomically to replay-state.json and intentionally survives topology reset.
- The main flow does not currently provide a production telemetry/alarm historian. MySQL and CSV are implemented for the separate dataset flow.

4. Gateway root architecture

The Gateway is the root of the LoRa MESH domain and the bridge to the site MQTT broker. It does not decrypt GLD payloads. It validates AppFrame framing, captures radio metrics, wraps the frame in JSON and publishes the result to MQTT.

Property	Current source value
Gateway ID	0x0001
Network interface	ESP32 Wi-Fi to site LAN
MESH radio	SX1262, 921.0 MHz, BW 125 kHz, SF9, CR 4/5
MESH sync / power	0x34 / 17 dBm
Status interval	10 seconds
MQTT retry	3 seconds
MQTT queue	8 volatile RAM entries, maximum 1024 bytes each

4.1 MESH receive processing

1	Receive raw LoRa frame and capture packet length, RSSI and SNR.
2	Decode AppFrame and reject frames explicitly addressed to a different provisionable Gateway ID.
3	Acknowledge CH_HELLO and answer CH_CONFIG_REQUEST when applicable.
4	Publish a JSON uplink wrapper and a separate topology event when applicable.
5	For alarm SENSOR_DATA, send the compact MESH ACK only when MQTT returned PublishedImmediately.

4.2 Gateway as durability boundary

An MQTT queue entry is not broker durability. If the broker is offline or publish fails, the frame can be retained in Gateway RAM for a short outage, but the Gateway deliberately withholds the alarm ACK. This causes the upstream CH chain to retain and retry the alarm.

FAILURE MODE

When the eight-entry queue is full, later uplink/topology messages are rejected and logged as queue drops. A larger queue would only extend the outage buffer; it would not solve broker connectivity, authentication or durable delivery.

5. Cluster Head architecture

Each CH is an active dual-radio router and local data manager. Radio A serves the GLD STAR domain. Radio B participates in the CH/Gateway MESH domain. The CH keeps GLD ciphertext opaque from sensor to server.

Resource	Capacity / behavior
NodeCache	32 latest GLD entries
Alarm queue	8 retained alarms
MESH TX queue	8 frames
Pending downlink store	16 GLD commands; one active lookup slot per node ID
Normal stale threshold	300,000 ms (5 minutes)
Cache expiration	3,600,000 ms (60 minutes)
Pending downlink TTL	1,800,000 ms (30 minutes) unless command overrides

5.1 CH state machine

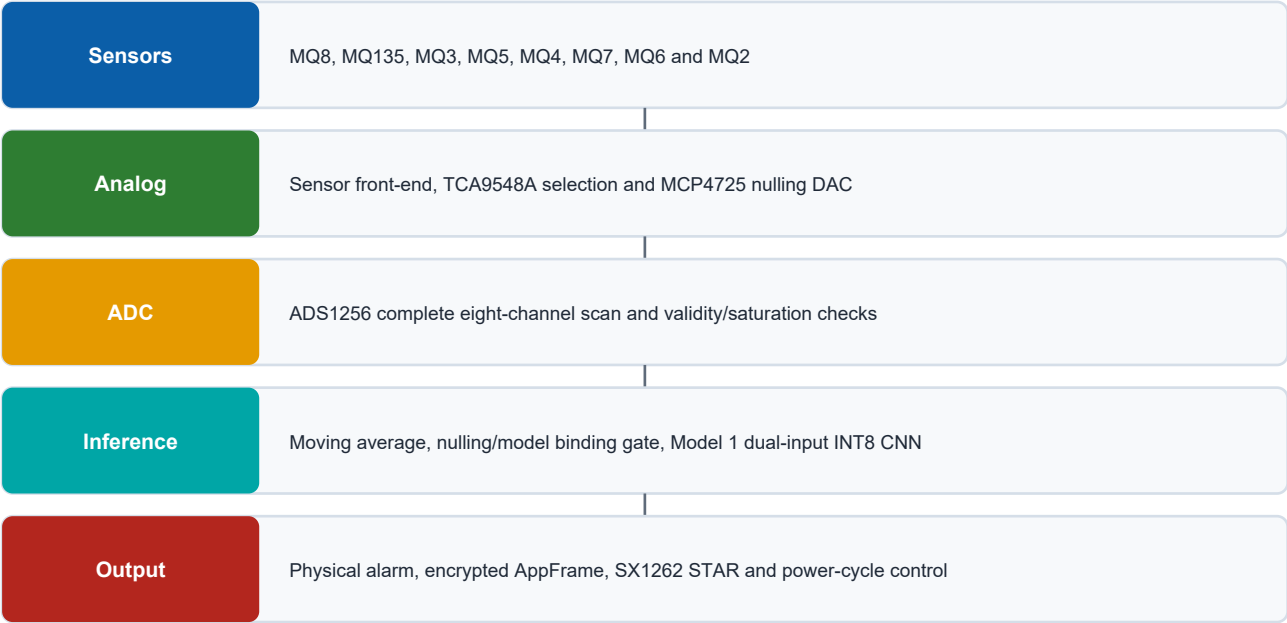
State	Responsibility
BOOT / WAIT_BATT	Initialize and wait for stable battery unless read-only field behavior is enabled
RADIO_INIT	Reset/init STAR and MESH radios; clear volatile caches and queues
JOINING	Broadcast CH_CONFIG_REQUEST, collect candidates and select parent
JOINED	STAR RX, MESH RX, TX drain, ACK tracking, HELLO, route verify and housekeeping
LOW_POWER	Continue receive processing; restrict TX according to battery threshold
PARENT_FAILOVER	Re-run discovery and retarget retained alarm traffic
RECOVERY	Controlled delay and ESP restart

5.2 Parent discovery and route stability

- CH broadcasts CH_CONFIG_REQUEST every 5 seconds during a 15-second joining window.
- Candidate selection primarily uses RSSI, with depth, SNR and ID tie-breakers; downstream candidates and obvious loops are rejected.
- Direct Gateway parent requires bidirectional quality. Alternate parent is retained for failover when applicable.
- HELLO interval is 300 seconds; parent health timeout is 960 seconds; route verification runs after 600 seconds plus up to 300 seconds jitter.

6. GLD edge architecture

GLD is the trusted sensing and cryptographic endpoint. It acquires all eight analog channels, applies the active nulling profile, validates the model binding, runs INT8 inference, drives local alarm outputs, encrypts the result and transmits it through STAR LoRa.



6.1 Active model contract

Item	Active Model 1
Profile	cnn-dualbranch-board-1-v1
Inputs	8 ADC channels plus 7 evidence features
Outputs	CO2/anomaly, Clean_Air, H2, LPG
Protocol mapping	{6, 0, 7, 1}
Production approval flag	true
Alarm decision	Any valid non-clear classification; confidence is reported but is not treated as calibrated percent LEL

INFERENCE GATE

Inference is valid only when all eight channels are valid, moving averages are primed, the model is initialized, the nulling profile is applied, and the persisted model/nulling binding matches the compiled model fingerprint.

6.2 Runtime modes

Mode	Connectivity	Behavior
inference	Wi-Fi/MQTT off; STAR on	Scan, CNN inference, alarm outputs and encrypted LoRa uplink
dataset	Wi-Fi/MQTT on	Validated eight-channel dataset records published directly to server topics
nulling	Wi-Fi/MQTT off	Local ADS/DAC calibration and NVS profile persistence

7. Wire protocol and cryptographic envelope

7.1 AppFrame

```

offset size field
0 1 magic = 0xAA
1 1 typeFlags
2 2 srcId (big-endian)
4 2 dstId (big-endian)
6 1 sequence
7 1 payloadLen
8 N payload
8+N 2 CRC16-CCITT-FALSE

```

Domain	Maximum payload	Maximum total AppFrame
STAR	64 bytes	74 bytes
MESH	80 bytes	90 bytes

7.2 GLD encrypted payload

Part	Size	Meaning
keyId	1 byte	Selects the provisioned AES key
nonce	12 bytes	Unique AES-GCM nonce
ciphertext	4 bytes	Encrypted gasClass, confidence and batteryMv
authentication tag	12 bytes	AES-GCM integrity and authenticity tag
total	29 bytes	Opaque payload relayed by CH and Gateway

AES-GCM additional authenticated data is nodeId + GLD sequence + record flags + keyId. This binds ciphertext to the logical GLD identity, sequence and alarm/external-power context.

7.3 GLDRecord inside a cluster response

```

nodeId:uint16BE + gldSeq:uint8 + flags:uint8 + payloadLen:uint8
+ encryptedPayload[29]
= 34 bytes per current record

cluster response header 6 bytes + 2 records x 34 bytes = 74 bytes

```

8. End-to-end message flows

8.1 Normal telemetry - pull model

1	GLD completes a valid scan and sends an encrypted SENSOR_DATA AppFrame to its configured CH.
2	CH validates destination and framing, then stores the latest opaque record in NodeCache. Normal data is not immediately pushed to the server.
3	Node-RED selects an installed route and publishes requestId plus hopList to gld/gateway/cmd/pull.
4	Gateway builds SERVER_PULL_REQUEST; intermediate CH nodes forward it according to hopList.
5	The final CH selects oldest normal, unsent and non-stale records. At most two current records fit in one response.
6	CLUSTER_DATA_RESPONSE travels upward through each active parent to Gateway.
7	Gateway publishes the raw frame wrapper. Node-RED correlates requestId, reconstructs GLD ownership, decrypts and publishes gld/server/decoded.

8.2 Alarm telemetry - immediate push

1	GLD marks any valid non-clear classification as alarm and creates a frozen encrypted frame. Battery-mode alarms are persisted in NVS.
2	CH accepts the record into NodeCache and alarm queue, then transmits an immediate alarm push toward its active parent.
3	Each relay serializes the alarm ACK transaction and retains the frame while waiting for its parent ACK.
4	Gateway publishes to MQTT. It ACKs the source CH only after immediate broker publish success, never for a merely queued volatile frame.
5	Node-RED verifies AES-GCM and replay state before publishing gld/server/alarm.

SECURE COMPLETION GAP

The compact STAR/MESH ACK format is CRC-protected but not cryptographically authenticated. Production GLD therefore rejects the compact ACK for clearing a persisted alarm. Server receipt and secure GLD-side alarm completion are not yet the same event.

9. Downlink and maintenance flows

9.1 Signed Server-to-GLD mode command

1	Operator/server publishes a high-level request to gld/server/cmd/node with cluster, node, command ID, TTL, mode and authorization.
2	Node-RED validates the authorization token and builds an eight-byte command: 0x81 + mode + commandId + four-byte AES-CMAC tag.
3	Gateway wraps the command as SERVER_NODE_COMMAND. It uses direct legacy delivery or routed v1 with an explicit unique hopList.
4	Intermediate CH nodes validate the previous hop and relay the unchanged payload. The final CH stores the pending downlink.
5	For external-powered GLD, CH can send immediately. For battery GLD, CH waits for the next uplink and sends inside the two-second RX window.
6	GLD verifies target, CMAC and monotonically newer command ID, persists replay state, stores the mode and restarts.

9.2 Routed command limits

Constraint	Current limit
Command bytes	8
Routed command encoding	29 CH hops under the 80-byte MESH payload
Pull encoding	39 CH hops under the 80-byte MESH payload
Node-RED automatic route walk guard	16 iterations
Command result acknowledgement	Not implemented end-to-end; MESH command remains fire-and-forget

9.3 Dataset path

Dataset mode deliberately bypasses CH and Gateway. GLD connects directly to the configured Wi-Fi and dataset MQTT broker.

```
GLD -> gas-leak-detector/<device>/dataset/data
-> Node-RED dataset validator
-> MySQL idempotency gate
-> CSV append

Control topics:
gas-leak-detector/<device>/dataset
gas-leak-detector/<device>/cmd/ack
gas-leak-detector/<device>/dataset/status
gas-leak-detector/<device>/dataset/summary
gas-leak-detector/<device>/nulling/result
```

A valid dataset row requires the canonical eight-sensor order, valid ADS status, finite non-saturated values, valid gain, applied nulling profile and valid encoding. Queue capacity inside GLD dataset mode is 16 records; a full queue stops the session rather than silently discarding additional rows.

10. Security and reliability model

Boundary	Protection	Residual limitation
GLD payload	AES-128-GCM; 12-byte nonce; 12-byte tag; AAD binds identity/sequence/flags	Requires correct per-device key provisioning and nonce uniqueness
AppFrame	Magic, strict length and CRC16-CCITT-FALSE	CRC detects corruption but is not authentication
CH/Gateway relay	Payload remains opaque; destination and capability checks	Routing/control frames are not all cryptographically authenticated
Server replay	Persistent nonce and encrypted fingerprint identity	Replay persistence path must be writable and operational
Downlink	Authorization token at server plus AES-CMAC command tag and GLD replay ID	No end-to-end command-result ACK
Alarm delivery	NVS frozen frame, CH retry/failover, Gateway ACK only after immediate MQTT publish	Compact RF ACK remains unauthenticated
MQTT transport	Supports credentials; generator can require TLS for remote host	Checked-in local snapshot is loopback without TLS; deployment state unverified

10.1 Failure containment

- GLD production transmission fails closed when no AES key is provisioned in NVS.
- CH rejects STAR traffic addressed to a different CH before cache, ACK, alarm, relay or downlink mutation.
- CH retains alarms across MESH retry and parent failover instead of treating radio TX success as server durability.
- Gateway distinguishes PublishedImmediately, QueuedVolatile and Rejected outcomes.
- Node-RED accepts legitimate sequence wrap/restart when nonce and ciphertext are fresh, but rejects exact encrypted replay and nonce reuse.

10.2 Observability

Each layer exposes distinct evidence. GLD logs sensor, inference, crypto and LoRa state. CH logs STAR receive, cache, parent selection, MESH TX and ACK state. Gateway logs Wi-Fi/MQTT connection, queue disposition and radio receipt. Node-RED exposes decoded, integrity, replay, topology and error topics. A complete proof should correlate all four layers for the same node, sequence and request or command ID.

11. Current configuration matrix

Area	Current source value	Operational note
GLD default identity	node 0x1001; target CH 0x0010	Provisioned values can override defaults through NVS
CH default identity	0x0010	Operator provisioning persists CH ID and resets cached parent state
Gateway root	0x0001	Must match CH root Gateway ID
STAR radio	920 MHz, BW125, SF7, CR4/5, sync 0x12, 17 dBm	GLD and CH Radio A must match
MESH radio	921 MHz, BW125, SF9, CR4/5, sync 0x34, 17 dBm	All CH Radio B and Gateway must match
GLD scan / TX	500 ms / 10 s	External-power inference cadence; battery uses one wake session
GLD RX window	2 s	CH battery downlink must arrive after GLD uplink
GLD battery session	30 s warm-up; 10 complete batches; up to 3 alarm attempts	Then DONE and CLR unless service hold is active
CH HELLO / health	300 s / 960 s	HELLO retry 3 with 1.5 s ACK timeout
Gateway MQTT queue	8 x 1024 bytes	Volatile outage buffer, not durable storage
Broker source default	port 1884; Gateway host placeholder	Deployment must provide the actual site host and secrets

DOCUMENT DRIFT

Several older final_design and README sections still contain previous IDs, versions, timeout values, confidence-threshold language or self-test-key behavior. This document prioritizes the current source snapshot over those older mirrors.

12. Production boundaries and unresolved gaps

Priority	Gap	Impact
High	Secure end-to-end alarm completion is unresolved	Production GLD cannot trust the current compact ACK to clear a persisted alarm
High	Gateway site broker host and credentials are deployment placeholders	Source alone cannot establish server connectivity
High	No production historian is visible in the main telemetry/alarm flow	Decoded events may depend on external consumers not represented in this repository
Medium	No end-to-end command result ACK	Server can submit a command but cannot prove that GLD executed it
Medium	Gateway MQTT queue is small and volatile	Extended outage can fill the queue and reject later messages
Medium	Automatic topology route builder has a 16-step guard	Wire format supports more hops than the current server route construction
Operational	Worktree contains uncommitted user changes	This PDF describes the current local snapshot, not necessarily a released Git commit

12.1 Evidence boundary

PROVEN IN THIS DOCUMENT

Static source values, frame formats, configured topics, queue sizes, routing behavior and checked-in flow structure.

NOT PROVEN IN THIS DOCUMENT

Firmware build success, flash contents, active NVS identities, live RF delivery, broker listener, MQTT CONNACK, deployed Node-RED revision, database availability, alarm consumer operation or physical sensor accuracy.

12.2 Recommended acceptance evidence

- Correlate one normal record: GLD LoRa TX -> CH cache -> routed pull -> Gateway MQTT publish -> Node-RED decrypt.
- Correlate one alarm record across every hop and verify the exact ACK disposition at Gateway.
- Verify broker listener, MQTT CONNACK and Node-RED deployed revision with production credentials hidden from logs.
- Verify the production consumer or historian for gld/server/decoded and gld/server/alarm.
- Record GLD model/nulling binding, active identity and AES key source without exposing the key.

Appendix A. Source map

The following repository files are the primary architecture evidence used for this document.

Area	Repository source
Version contract	firmware/shared/include/FirmwareVersion.h
AppFrame and message constants	firmware/shared/include/ProtocolConstants.h; AppFrame.h; shared/src/AppFrame.cpp
GLD configuration and runtime	firmware/config/GldConfig.h; firmware/gld/src/GldUnifiedMain.cpp
GLD power	firmware/gld/include/GldPower.h; firmware/gld/src/GldPower.cpp
Active model	firmware/gld/models/model_1/ModelMetadata.h; NeuralNetwork.cpp
CH configuration and runtime	firmware/config/ChConfig.h; firmware/ch/src/ChStarMeshRuntimeMain.cpp
Gateway configuration and runtime	firmware/config/GwConfig.h; firmware/gateway/src/GatewayMqttMeshMain.cpp
Server/broker configuration	firmware/config/ServerConfig.h
Node-RED decode	server/nodered/functions/pertamina-gld-decode.js
Node-RED command signing	server/nodered/functions/pertamina-gld-build-command.js
Main server flow	server/nodered/pertamina-gld-server.flow.json
Dataset flow	server/nodered/pertamina-gld-dataset.flow.json

A.1 Message type reference

Value	Name	Direction / use
0x10	MSG_SENSOR_DATA	GLD uplink and alarm/record push
0x14	MSG_NODE_DOWNLINK	Final CH to GLD command
0x30	MSG_SERVER_PULL_REQUEST	Gateway/server toward target CH
0x31	MSG_CLUSTER_DATA_RESPONSE	Target CH response toward Gateway
0x32	MSG_SERVER_NODE_COMMAND	Gateway/server command toward owning CH
0x33	MSG_CH_HELLO	CH topology announcement
0x34	MSG_CH_CONFIG_REQUEST	CH discovery broadcast
0x35	MSG_CH_CONFIG_RESPONSE	CH/Gateway discovery response
0x36	MSG_CH_HELLO_ACK	Parent/Gateway acknowledgement of HELLO

End of document